Case (ii):

$P = 2$ MW; $R = 40\ \Omega$; $V = 100$ kV

$$\therefore \text{Current, } I = \frac{P}{V}$$

$$= \frac{2\times10^6}{100\times10^3} = 20A$$

Power loss $= I^2R$

$$= (20)^2 \times 40 = 0.016\times10^6 W$$

$$\% \text{ of power loss} = \frac{0.016\times10^6}{2\times10^6}\times100\%$$

$$= 0.008\times100\% = 0.8\%$$

Thus it is clear that when an electric power is transmitted at higher voltage, the power loss is reduced to a large extent.

EXAMPLE 4.16

An ideal transformer has 460 and 40,000 turns in the primary and secondary coils respectively. Find the voltage developed per turn of the secondary if the transformer is connected to a 230 V AC mains. The secondary is given to a load of resistance $10^4\Omega$. Calculate the power delivered to the load.

Solution

$N_P = 460$ turns; $N_S = 40{,}000$ turns

$V_P = 230$ V; $R_S = 10^4\ \Omega$

(i) Secondary voltage,

$$V_S = \frac{V_P N_S}{N_P} = \frac{230\times40{,}000}{460}$$

$$= 20{,}000V$$

Secondary voltage per turn, $\frac{V_S}{N_S} = \frac{20{,}000}{40{,}000}$

$$= 0.5\ V$$

(ii) Power delivered

$$= V_S I_S = \frac{V_S^2}{R_S} = \frac{20{,}000\times20{,}000}{10^4} = 40\ kW$$

EXAMPLE 4.17

An inverter is common electrical device which we use in our homes. When there is no power in our house, inverter gives AC power to run a few electronic appliances like fan or light. An inverter has inbuilt step-up transformer which converts 12 V AC to 240 V AC. The primary coil has 100 turns and the inverter delivers 50 mA to the external circuit. Find the number of turns in the secondary and the primary current.

Solution

$V_p = 12$ V; $V_s = 240$ V

$I_s = 50$mA; $N_p = 100$ turns

$$\frac{V_S}{V_P} = \frac{N_S}{N_P} = \frac{I_P}{I_S} = K$$

Transformation ratio, $K = \frac{240}{12} = 20$

The number of turns in the secondary

$$N_S = N_P \times K = 100\times20 = 2000$$

Primary current,

$I_p = K \times I_s = 20 \times 50$ mA $= 1$ A

4.7 ALTERNATING CURRENT

4.7.1 Introduction

In section 4.5, we have seen that when the orientation of the coil with the magnetic field is changed, an alternating emf is induced and hence an alternating current flows in the closed circuit. **An alternating voltage is the voltage which changes polarity at regular intervals of time and the direction of the resulting alternating current also changes accordingly.**

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